

Figure 7: R file location

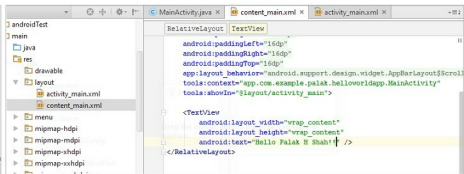


Figure 8: 'Hello World' code file

4. **AndroidManifest.xml**: This is the manifest file that explains all the basics of the app and its components.
5. **res/values**: This is the directory that stores various XML files, which contain a collection of resources like the string and colour definitions.
6. **String file**: This contains all the text used by the app—for example, labels for buttons, the default text, etc. It is located in the **res/values** folder.
7. **R file**: This fills the gap between the activity Java files and the resources. It is generated automatically.

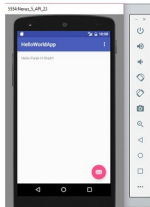


Figure 9: The 'Hello World' app emulator

Running the 'Hello World' app

By default, the project will have the text 'Hello World'. If you wish to modify the text, you can do so in the **content_main.xml** file.

To run the app, click on the **Play** button and select the AVD to play the app. The **AVD manager** will launch the AVD and our app will be hosted on it. You can see the output in Figure 9.

Application components

Activities: An activity is the component that provides a screen to interact with. Each activity has its own window to host the user interface.

An app can contain more than one activity, which is integrated with others. There is always a main activity, which is being executed while launching the app. Then, any activity can initiate other activities to execute different tasks.

To create an activity, you need to create a subclass of activity. In your class, you should implement methods of the activity life cycle such as when the activity is created, destroyed, stopped and resumed.

Given below are the important callback methods.

- **onCreate()**: This method is called while creating the activity. We must implement this method, as only this

will initialise all other components of the app. To define the layout of the app, we should call **setContentView()** here so that while running the app, it will open with the proper layout.

- **onPause()**: This method is called when the activity becomes invisible to the user and another activity gets priority.
 - **onRestart()**: This is called when the activity starts its execution again.
 - **onStart()**: This is called once the activity makes itself visible to users.
 - **onResume()**: This is called once the activity resumes its execution.
 - **onStop()**: This is called once the activity is no longer visible.
 - **onDestroy()**: This is called to finish executing the activity.
- The diagram in Figure 10 explains the life cycle of an activity during app execution.

Intent: Android Intent is the message/value passed between app components such as activities, services, content providers, etc. It is used to request an action from another component. It is used with the method **startActivity()** to initialise the action.

The main uses of Android Intents are:

1. To start the service
 2. To launch the activity
 3. To display any URL
 4. To dial a call
 5. To display a list of contacts, etc
- There are two types of Android Intents.
- 1) **Implicit Intent**: This specifies the general action, which is taken care of by any available component instead of specifying the component to handle the action. For example, you may write the following code to view the Google home page:

```
Intent intent1=new Intent();
intent1.setAction(Intent.ACTION_VIEW);
intent1.setData(Uri.parse("http://www.google.com"));
startActivity(intent1);
```

- 2) **Explicit Intent**: In contrast, this specifies the component.